

2 Barley is often grown in soils that are high in salt (sodium chloride).

Researchers tested the salt tolerance of two varieties of barley, **X** and **Y**, by germinating grains in seven different concentrations of salt solution.

- Seven sets of barley grains of each variety were soaked in the different concentrations of salt solution for 12 hours. Each set contained 50 grains.
- Each set of grains was then split into groups of 10 grains. These were placed in Petri dishes on filter paper soaked in the same concentrations of salt solution that were used for soaking the grains.
- Lids were placed on the Petri dishes and the grains were left at 20 °C to germinate.

Fig. 2.1 shows the arrangement of the grains in the Petri dishes.

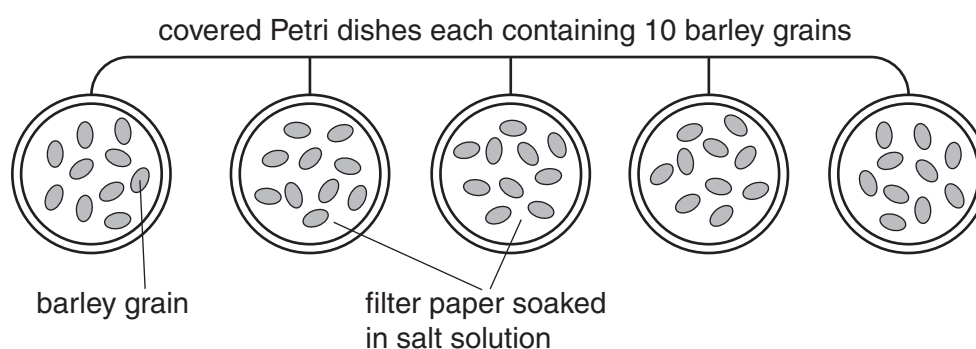


Fig. 2.1

The grains were then observed at 8 hour intervals for a total of five days. The percentage of grains in each set of 50 that had germinated each day was calculated.

The first appearance of the radicle (root) was used to indicate that the grains had germinated.

If no radicle appeared after five days, germination was considered to have failed.

The researchers repeated the whole investigation several times and calculated mean results.

(a) (i) Describe **one** variable that has been standardised in this investigation.

.....
..... [1]

(ii) Suggest **one** other variable that should be standardised.

.....
..... [1]

Table 2.1 shows the results of this investigation.

Table 2.1

concentration of salt solution /mmol dm ⁻³	mean (cumulative) percentage of grains that had germinated each day			
	1		5	
	X	Y	X	Y
0	93.8	92.0	98.0	97.0
10	93.0	90.2	97.4	96.4
20	92.2	87.0	96.6	93.6
30	96.4	90.0	96.4	93.0
40	92.0	90.4	95.4	92.4
50	91.3	91.0	95.0	91.6
60	91.2	90.0	95.0	91.0

- (b) (i) Identify **two** results which may be anomalous.
Show your answer by circling the two values. [2]
- (ii) State a statistical test that could be used to find out if the correlation between the percentage germination and the salt concentration is significant.
Give a reason for your answer.

test

reason

.....

[2]

- (c) Use the data from Table 2.1 to state **three** conclusions that are valid for **both** varieties, **X** and **Y**.

..... [3]

- (d) The researchers concluded that variety **X** would be better suited than variety **Y** to grow in salty soil.

Other researchers did not think there was sufficient evidence to make definite conclusions.

- (i) State the evidence in Table 2.1 that supports the conclusion that variety **X** is better suited than variety **Y** to grow in salty soil.

.....
..... [1]

- (ii) The researchers decided to carry out further investigations to find out which is the most suitable variety, **X** or **Y**, to grow in salty soils.

Suggest **one** other investigation they could carry out.

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..... [1]

[Total: 11]